

Expanding the Specification Space for FIR Filter Designs with AI

Andrei Cozma, Channing Tan

Abstract—The Parks-McClellan algorithm produces optimal equiripple FIR filters efficiently and is well-understood theoretically. However, it has fundamental design constraints and practical failure modes that motivate alternative approaches. This paper presents two complementary methods: feasibility prediction and direct optimization. Feasibility prediction uses machine learning classifiers to determine whether a specification is achievable before running the algorithm, with XGBoost achieving 95 % average accuracy across filter types. Direct optimization bypasses Parks-McClellan entirely, using gradient descent with differentiable loss functions to find filter coefficients directly. This approach naturally accommodates alternative objectives such as mean-squared error, dB-scale penalties, and constraint-based formulations.

Index Terms—FIR Filter, Parks-McClellan, Machine Learning, Feasibility Prediction, Gradient Optimization

I. INTRODUCTION

DIGITAL filters have been a continued source of interest for electrical engineers and computer scientists for many decades. They help eliminate noise in signals to isolate the main component. Digital filters can be divided into four main types: low-pass (LP), high-pass (HP), band-pass (BP), or band-stop (BS). These each correspond to an area of the frequency spectrum that noise is removed and signal is passed. Furthermore, digital filters can also be categorized as Finite Impulse Response (FIR) or Infinite Impulse Response (IIR) filters. For communication systems, FIR filters are preferred due to their stability and linearity [1].

Machine learning is currently a prominent topic for its capability to optimize a variety of functions. For classification algorithms, Random Forests (RF), Extreme Gradient Boosting (XGBoost), and Support Vector Machines (SVMs) are popular. To train deep neural networks, backpropagation and gradient descent are used.

To design FIR filters, the Parks-McClellan Algorithm [2] (PM) is a widely used method for designing optimal equiripple FIR filters. Given a fixed filter length, it minimizes the maximum weighted approximation error, producing equiripple behavior where error oscillates with equal magnitude throughout passband and stopband. However, many popular implementations suffer from robustness issues [3]. These failures stem from algorithmic vulnerabilities including ill-conditioned numerical interpolation at high filter orders and sensitivity

to initial extremal frequency placement [3]. For the SciPy library, failure manifests as non-convergence (iterations fail to stabilize) or numerical failures (NaN/inf coefficients). Because there is no feasibility check before running PM, time may be wasted on specifications that are simply not feasible. In addition, quantization can undo optimal design. To address the challenges mentioned above, this paper proposes two solutions using modern technology. The first is to have a feasibility check with AI. The second is to utilize gradient descent and backpropagation to directly optimize coefficients. The organization of this paper is as follows. Section II identifies the relationship between specifications and failures in PM. Section III gives an overview of models used for feasibility analysis. Section IV details the methodology used for training a replacement for PM. Section V details and analyzes simulation results. Section VI provides the conclusion.

II. DESCRIPTION OF SPECIFICATIONS

There are five specifications that are considered in this paper. Filter order refers to the number of coefficients that should be generated for any given filter. Transition width refers to width between the passband and stopband frequencies. The passband is specified with a cutoff frequency for LP and HP filters or bandwidth and center frequency for BP and BS filters. To use `scipy.signal.remez`, max frequencies are not to exceed half the sampling frequency. Assuming a default frequency of Hz, this means all frequencies must be less than 0.5 Hz. The range of specifications and their distributions are listed in Table I.

TABLE I
SPECIFICATIONS CONSIDERED

Parameter	Range	Distribution
Filter Order	21–1001	Uniform
Transition Width	0.01–0.12	Log-Uniform
Cutoff (LP/HP)	0.03–0.45	Uniform
Bandwidth (BP/BS)	0.05–0.35	Log-Uniform
Center Frequency (BP/BS)	0.15–0.35	Uniform

The required filter order to meet specifications can be estimated as:

$$M \approx \frac{-10 \log_{10}(\delta_1 \delta_2) - 13}{2.324(\omega_s - \omega_p)} \quad (1)$$

where δ_1 and δ_2 are passband and stopband tolerances, and $(\omega_s - \omega_p)$ is transition width in normalized frequency. This

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formula reveals the fundamental trade-off: shorter filters, narrower transitions, and tighter tolerances cannot be achieved simultaneously.

Once specifications were identified, around 50,000 samples of data were randomly generated and basic relationships between features and success were found. Fig. 1 shows the correlation between features. As shown, order and success are highly correlated.

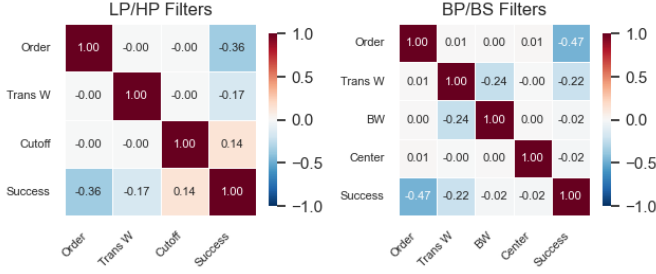


Fig. 1. Correlation matrix showing relationships between filter specification parameters and design success. Filter order shows strong positive correlation with success rate.

This idea is reinforced by the feasibility heatmap shown in Fig. 2. As illustrated, failure rate increases with higher filter order and narrower transition width.

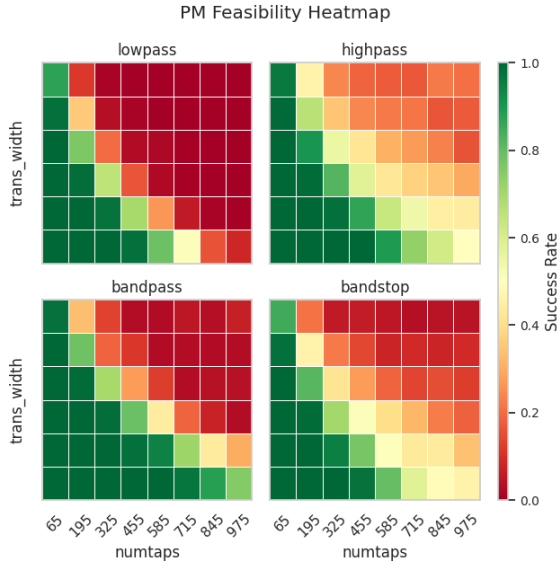


Fig. 2. Feasibility heatmap showing Parks-McClellan failure rate as a function of filter order (vertical) and transition width (horizontal). Darker regions indicate higher failure rates, concentrated in high-order, narrow-transition specifications.

Finally, the relationship between filter type and success rate was found, as shown in Table II. Band-pass filters achieve the highest success rate, while band-stop and high-pass filters have the lowest rates.

TABLE II
SUCCESS RATE BY FILTER TYPE

Filter Type	Success Rate
Band-Stop	31.3%
High-Pass	34.6%
Low-Pass	36.0%
Band-Pass	54.3%

III. DESCRIPTION OF FEASIBILITY ANALYSIS MODELS

The Parks-McClellan algorithm provides no indication of whether a given specification is achievable at a particular filter order until after execution completes. Current practice requires trial-and-error: attempt a design, evaluate the result, increment order, and repeat until specifications are met or deemed infeasible. This wastes computation on impossible targets and provides no guidance on which parameters to adjust.

Feasibility analysis addresses this gap by predicting whether a (specification, order) pair will succeed before running the algorithm. Each model was given 50,000 different samples of filter specifications. Since the 50,000 samples were split among the four types of filters, each model handled 12,500 filters. While it is possible to have a single model learn all the features for the different types of filter, having one model for each filter was seen as more effective since the characteristics of each filter varied widely. To train each model, a train-test split of 80-20 of the data was used. Descriptions of each model are described below.

A. Random Forest

Random Forests are an ensemble model made from growing decision trees with random selection of features [4]. By averaging multiple trees, the variance is decreased, leading to less overfitting.

B. SVM

Support Vector Machines (SVMs) [6], [7] are a class of supervised machine learning algorithms widely utilized for classification and regression tasks. The core principle of SVM is to find a hyperplane that best separates data points into distinct classes, while maximizing the margin between the classes. The margin is defined as the distance between the hyperplane and the closest data points from each class, known as support vectors.

C. XGBoost

XGBoost [5] is a gradient boosting algorithm, widely recognized for its superior performance in machine learning competitions and real-world applications. It belongs to the family of ensemble methods, which combine the predictions of multiple weak models to produce a stronger model. At its core, XGBoost builds an ensemble of decision trees in a stage-wise fashion, where each successive tree corrects the errors of its predecessor. The key innovation behind XGBoost lies in its optimization process, which minimizes the loss function using

both first-order and second-order gradient information. This leads to faster convergence and better performance compared to traditional gradient boosting methods.

While feasibility prediction helps avoid wasted computation on infeasible specifications, it does not address the limitations of Parks-McClellan itself. The following section presents an alternative approach that bypasses the algorithm entirely.

IV. DIRECT OPTIMIZATION

Direct optimization approaches the filter design problem by optimizing the filter coefficients themselves through gradient descent, rather than using the Parks-McClellan iterative exchange algorithm. The key distinction is that there is no neural network involved: the learnable parameters are the filter tap values directly. This approach enables optimization of objectives that Parks-McClellan cannot natively handle, such as multi-objective criteria, sparsity constraints, and quantization-aware design.

To compute differentiable loss functions, the frequency response must be evaluated from the current coefficient estimates. This implementation uses Fast Fourier Transform based evaluation, which provides efficient direct computation of the frequency domain representation. The magnitude response at each frequency bin can then be compared against specifications to form optimization objectives.

A. Loss Function Formulations

Two fundamentally different objectives exist for filter optimization: minimizing mean error (average case) or minimizing maximum error (worst case). Parks-McClellan uses the Chebyshev criterion, which minimizes maximum weighted error and produces equiripple behavior. Mean-based objectives are easier to optimize but yield different filter characteristics. We propose four loss formulations spanning both paradigms to enable systematic comparison of their optimization behavior, convergence properties, and resulting filter quality. Let H_p and H_s denote passband and stopband magnitude responses, with $\overline{(\cdot)}$ denoting mean over frequency and weights $W = 1/\delta$ following Parks-McClellan convention.

1) *Mean-Based Losses*: The weighted L2 loss minimizes mean squared deviation:

$$\mathcal{L}_{L2} = W_p \cdot \overline{|H_p - 1|^2} + W_s \cdot \overline{|H_s|^2} \quad (2)$$

This is identical to least-squares regression over frequency points. Advantages include smooth convex optimization, stable convergence, and no hyperparameters. However, it allows large localized errors if the average is low, and will not produce equiripple behavior since it targets a fundamentally different objective than Parks-McClellan.

The dB-scale loss applies the same mean-squared objective in logarithmic scale:

$$\mathcal{L}_{dB} = W_p \cdot \overline{(20 \log_{10} |H_p|)^2} + W_s \cdot \overline{\text{ReLU}(20 \log_{10} |H_s| - A_t)^2} \quad (3)$$

where A_t is target stopband attenuation. This addresses the vanishing gradient problem: at -60 dB the linear magnitude

is 0.001, making gradients negligible, while in dB scale gradients remain meaningful. The asymmetric treatment (passband penalizes any deviation, stopband only penalizes insufficient attenuation) reflects that exceeding stopband requirements is acceptable. Despite the transform, this remains mean-based and will not produce equiripple.

2) *Max-Based Losses*: The soft minimax loss approximates the Chebyshev criterion:

$$\mathcal{L}_{mm} = \frac{1}{T} \log \sum \exp(TW_p |H_p - 1|) + \frac{1}{T} \log \sum \exp(TW_s |H_s|) \quad (4)$$

As $T \rightarrow \infty$, this approaches $\max(W_p |H_p - 1|) + \max(W_s |H_s|)$, matching Parks-McClellan. The formulation is related to softmax in classification and the Boltzmann distribution, where T is a temperature parameter. This is theoretically the correct objective for equiripple behavior. However, the worst point dominates optimization, and temperature presents a tradeoff: low T gives poor max approximation while high T causes gradient instability.

The constraint-based loss enforces tolerance requirements:

$$\mathcal{L}_{con} = W_p \cdot \text{ReLU}(\max |H_p - 1| - \delta_p) + W_s \cdot \text{ReLU}(\max |H_s| - \delta_s) \quad (5)$$

The loss is zero when specifications are satisfied, matching how filter specs are typically written. This produces sparse gradients, focusing effort only on violated constraints. Disadvantages include slow initial progress due to sparse gradients and linear growth with violation magnitude, providing weaker pull toward satisfaction than quadratic penalties. Table III summarizes the key properties of each loss function.

TABLE III
LOSS FUNCTION COMPARISON

Property	L2	dB	Minimax	Constraint
Objective	Mean	Mean	Max	Max
Scale	Linear	Log	Linear	Linear
PM-like	No	No	Yes	Partial
Gradients	Smooth	Smooth	T-dep.	Sparse

B. Coefficient Parameterization

Two parameterization modes determine how coefficients are represented during optimization. In the half-coefficient mode, only half of the filter coefficients are optimized as free parameters. The complete filter is constructed by mirroring these values to enforce perfect symmetry. This guarantees linear phase response by construction and reduces the parameter count by approximately half. The symmetry constraint cannot be violated during optimization.

In the full coefficient mode, all filter coefficients are independent optimization variables. Linear phase is not guaranteed and must be measured post-optimization if required. This allows exploration of non-linear phase designs and provides greater flexibility, at the cost of increased parameter count and no structural phase constraints.

C. Optimization Challenges

Unlike supervised learning, the primary risks are optimization failures rather than generalization failures. Since each optimization produces a single filter for a single specification, there is no overfitting concern and no need for validation sets. The loss function directly represents the true objective.

However, gradient-based optimization on difficult specifications can exhibit several failure modes. Divergence occurs when gradients become explosive, with gradient norms exceeding tens of thousands observed on tight tolerance specifications. Oscillation manifests as loss values that bounce without making progress toward convergence. Premature termination happens when iteration limits are reached before specifications are satisfied. These challenges require careful learning rate selection, gradient clipping strategies, and sufficient iteration budgets. Different loss formulations exhibit different stability characteristics across the specification space.

Having established both approaches, we now evaluate their performance on a diverse set of filter specifications.

V. SIMULATION RESULTS

A. Direct Optimization Results

Direct optimization was evaluated using the weighted L2 loss with half-coefficient parameterization to enforce linear phase by construction. Fig. 3 shows a representative stopband frequency response achieved through this approach. Fig. 4 compares the distribution of passband ripple between Parks-McClellan and direct optimization across the test specifications. The direct optimization approach exhibits higher ripple on average, reflecting the fundamental difference between mean-based (L2) and max-based (Chebyshev) objectives discussed in Section IV.

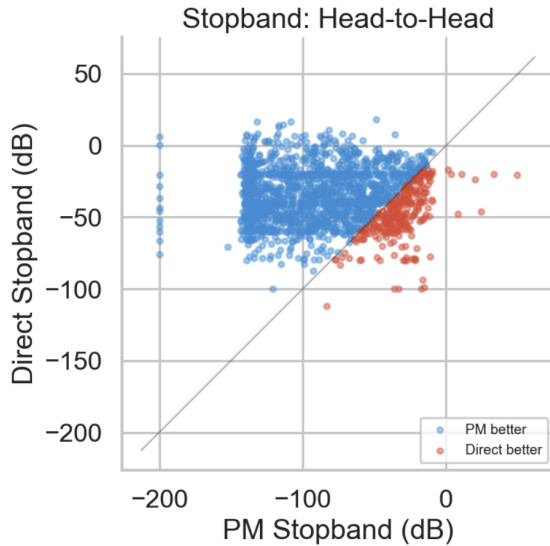


Fig. 3. Representative stopband frequency response from direct optimization using weighted L2 loss. The response shows smooth attenuation characteristic of mean-based optimization rather than the equiripple behavior of Parks-McClellan.

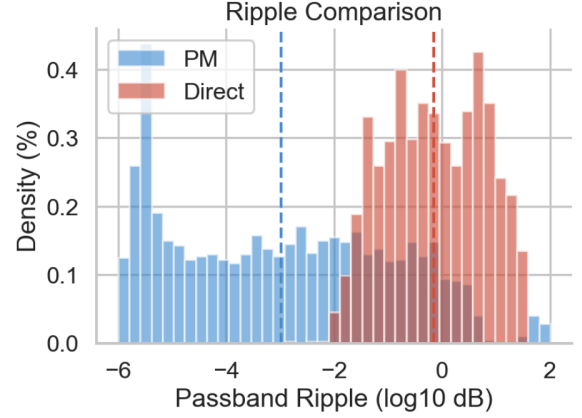


Fig. 4. Passband ripple distribution comparing Parks-McClellan (left) and direct optimization (right). Direct optimization exhibits higher and more variable ripple due to the mean-based L2 objective, which tolerates localized deviations that Parks-McClellan's minimax criterion would penalize.

B. Feasibility Model Results

Table IV summarizes classification performance across all models and filter types. XGBoost achieved the highest average scores across all metrics, followed closely by Random Forest. SVM underperformed relative to the tree-based methods, particularly on low-pass and band-pass filters. Bold values indicate the best result for each filter type within each metric.

TABLE IV
FEASIBILITY PREDICTION RESULTS BY MODEL AND FILTER TYPE

Model	Filter	Acc.	Prec.	Rec.	F1
SVM	LP	0.88	0.87	0.77	0.82
	BP	0.79	0.81	0.80	0.80
	HP	0.83	0.83	0.89	0.86
	BS	0.81	0.80	0.84	0.82
	Average	0.83	0.83	0.83	0.83
RF	LP	0.96	0.96	0.94	0.95
	BP	0.93	0.94	0.93	0.94
	HP	0.94	0.93	0.93	0.94
	BS	0.94	0.95	0.94	0.94
	Average	0.94	0.94	0.95	0.94
XGB	LP	0.97	0.96	0.94	0.95
	BP	0.93	0.95	0.92	0.95
	HP	0.94	0.94	0.95	0.95
	BS	0.95	0.94	0.96	0.95
	Average	0.95	0.95	0.94	0.95

Confusion matrix analysis reveals that classifier performance correlates with class balance in the test set. XGBoost achieves the highest accuracy on lowpass filters (97%), with band-stop, high-pass, and low-pass specifications all having similar success rates (31–36% from Table II). Band-pass filters, with the highest success rate (54.3%), show slightly lower but still strong classification accuracy (93%).

Fig. 5 shows ROC curves stratified by filter type and order quartile. Lowpass and bandpass filters achieve near-perfect discrimination ($AUC > 0.97$) across all quartiles. Highpass filters show degraded performance in the third quartile (AUC

= 0.881), corresponding to moderate filter orders where the feasibility boundary is most ambiguous.

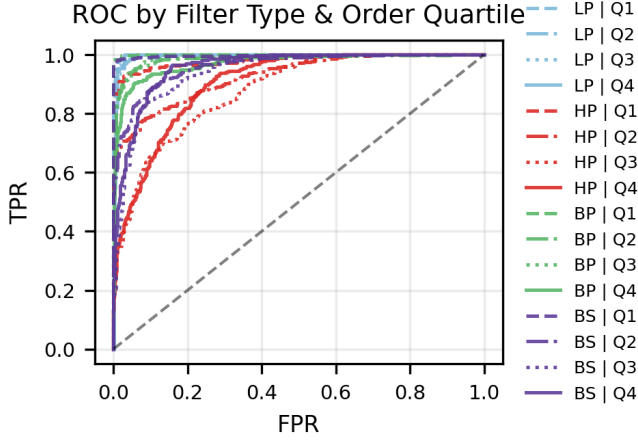


Fig. 5. ROC curves stratified by filter type and order quartile. Performance varies significantly across filter types, with lowpass and bandpass achieving consistently high AUC while highpass shows degraded discrimination at moderate orders.

VI. CONCLUSION

This paper presented two complementary approaches to address Parks-McClellan limitations. Feasibility prediction using machine learning achieves 95% average accuracy with XGBoost, enabling pre-screening of specifications before algorithm execution. Direct optimization bypasses the algorithm entirely, using gradient descent with differentiable loss functions to optimize filter coefficients directly.

Parks-McClellan remains the preferred method when linear phase is required and specifications are known to be feasible. The approaches presented here address scenarios where feasibility is uncertain or where alternative optimization objectives are desired.

Both approaches have limitations. The feasibility classifiers, while accurate, still misclassify 5% of specifications on average, which can waste computation on predicted-feasible but actually-infeasible designs. Direct optimization produces higher passband ripple than Parks-McClellan because mean-based losses do not target equiripple behavior. The soft minimax loss theoretically matches the Chebyshev criterion but presents gradient stability challenges at high temperature values.

Future work includes systematic ablation studies comparing the four proposed loss formulations across diverse specifications to characterize their convergence properties, stability regimes, and trade-offs between optimization difficulty and filter quality. Additional directions include amortized neural networks that predict filter coefficients in a single forward pass rather than per-specification optimization, hybrid approaches combining Parks-McClellan initialization with gradient refinement, and quantization-aware training to maintain filter quality after fixed-point implementation.

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